[INCH-POUND]
A–A–59789/3A
21 July 2010
SUPERSEDING
A–A–59789/3
26 March 2006

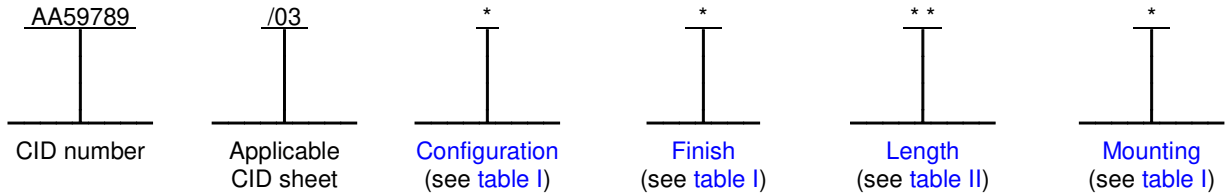
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

# HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 5 PIECE, FOR COLD PLATE APPLICATIONS, .225 X .225 INCH BODY SIZE, WITH TWO MOUNTING HOLES, WITH SCREW ACTUATION AND RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59789.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).




Example: AA59789/03NH60T is the PIN for a hard black anodize finished, 5.8 inch (147.3 mm) long card holder with a retained actuating screw. The card holder also feature two tapped mounting holes for use with 0–80 UNF 2B fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in place providing high resistance to shock and vibration while providing maximum thermal transfer.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and meet the general requirements specified in CID A–A–59789. Card holders shall have a screw retention feature to prevent unintentional disassembly of the card holder.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59789.

Actuating screw hex drive socket. The dimension for hex drive socket shall be .094 inch (2.38 mm) across flats for mounting options "S", "R", "P", "U", and "T".

Cold plate slot width. The recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59789).

Installation torque. The recommended nominal installation torque is as follows: 6 inch-pounds (0.68 N-m) for assemblies of configuration "A", "C", "F", or "W" and 7.2 inch-pounds (0.81 N-m) for assemblies of configurations "D", "J", "L", or "N".



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Configuration. The configuration of a card holder shall be as specified in table I. The details of a particular configuration consist of those on figures 1 and 2, and may include those on figures 3 and 4.

TABLE I. Configurations.

| Configuration | Mounting body style | Finish options | Mounting options | Applicable figures | Hardware options |
|---|---|---|---|---|---|
| A | A | B, C, H, P | S, R, T, P | 1 and 2 | No hardware options added |
| C | C | B, C, E, H | S, R, T, U | | |
| F | A | B, C, H, P | S, R, T, P | 1, 2, and 3 | Lockwasher and flat washer |
| W | C | B, C, E, H | S, R, T, U | | |
| N | A | B, C, H, P | S, R, T, P | 1, 2, and 4 | Screw self-lock element |
| L | C | B, C, E, H | S, R, T, U | | |
| J | A | B, C, H, P | S, R, T, P | 1, 2, 3, and 4 | Lockwasher, flat washer, and screw self-lock element |
| D | C | B, C, E, H | S, R, T, U | | |

Finish. The finish designator shall be as specified in A–A–59789. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "E" (electroless nickel), "P" (electrodeposited nickel), or "H" (hard black anodize). The center mounting wedge is not finished for card holders using mounting option "P".

Length, expanded and relaxed dimensions. The length designator shall be as specified in A–A–59789 and the lengths available for this CID specification sheet are listed in table II. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

TABLE II. Additional assembly dimensions (see figure 1). 1/

| PIN length designator | Dimension "L" ±.02 (0.5) | Mounting body style | Dimension "A" ±.01 (0.3) | Dimension "B" ±.02 (0.5) | Dimension "C" Minimum |
|---|---|---|---|---|---|
| 50 | 4.8 (121.9) | A | 1.50 (38.1) | .30 (7.6) | .275 (6.99) |
| | | C | 1.50 (38.1) | .30 (7.6) | .275 (6.99) |
| 55 | 5.3 (134.6) | A | 2.00 (50.8) | .55 (14.0) | .275 (6.99) |
| | | C | 1.50 (38.1) | .30 (7.6) | .275 (6.99) |
| 60 | 5.8 (147.3) | A | 2.50 (63.5) | .80 (20.3) | .275 (6.99) |
| | | C | 1.50 (38.1) | .30 (7.6) | .275 (6.99) |
| 65 | 6.3 (160.0) | A | 3.00 (71.2) | 1.05 (26.7) | .275 (6.99) |
| | | C | 1.50 (38.1) | .30 (7.6) | .275 (6.99) |

1/ Dimensions are in inches. Millimeters are given for information only.

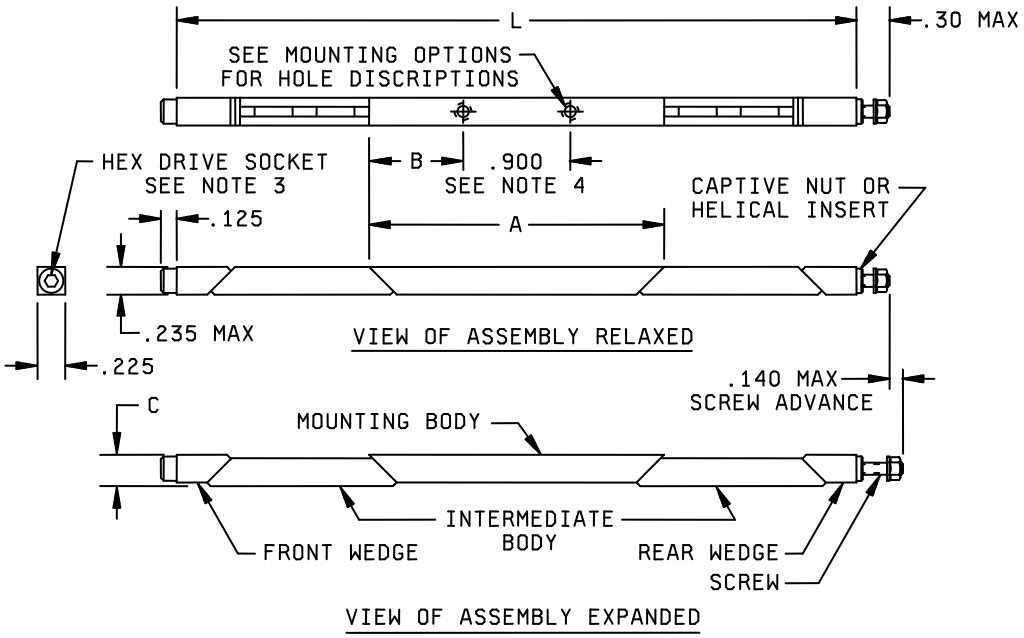
Mounting. The mounting designators shall be as specified in A–A–59789. The mounting options available for this CID specification sheet are as follows: "S" (tapped 2-56 holes), "R" (rivet mount holes with counterbore and countersink), "T" (tapped 0-80 holes), "P" (Ø.062 x .040 index pins), or "U" (tapped metric M2 x 0.4 holes). See figure 1 for mounting hole spacing requirements. Card holders using mounting option "P" and "R" may be shipped unassembled. The center mounting wedge is not finished for card holders using mounting option "P".



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Rivet mounting holes. The holes used for rivet mounting shall be .066/.073 inch (1.68/1.85 mm) diameter, countersunk 100 degrees by .140 inch (3.56 mm) diameter through holes with a counterbore of .156 inch (3.96 mm) diameter with a depth of .165 inch (4.19 mm) for style C mounting bodies and .175 inch (4.45 mm) for style A mounting bodies.

Rivets. This card holder uses rivet type A as specified in A–A–59789 when rivet mounting is used.




| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .125 | 3.18 | .225 | 5.72 | .30 | 7.6 |
| .140 | 3.56 | .235 | 5.97 | .900 | 22.86 |

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm).
4. Tolerance for the hole spacing is ±.010 inch (0.25 mm) for mounting body style A and ±.005 inch (0.51 mm) for mounting body style C.

FIGURE 1. Relaxed and expanded dimensions.



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Screw retention locknut (see figure 2). Card holders shall have a screw retention locknut intended to prevent unintentional disassembly of the card holder. The locknut shall be fabricated of a corrosion resistant material.

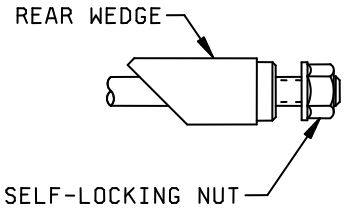



FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

Hardware options. Card holders can have the following options: no hardware options added, lockwasher and flat washer (see figure 3), screw self lock element (see figure 4), or lockwasher, flat washer, and screw self lock element (see figures 3 and 4). See table I for the correct PIN configuration identifier.

No hardware options added. The base configuration of the card holder is as depicted by figures 1 and 2 . Card holders not requiring any hardware options shall include configuration identifier "A" or "C" in the PIN (see classification, notes, and table I).

Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include configuration identifier "F" or "W" in the PIN (see classification, notes, and table I).

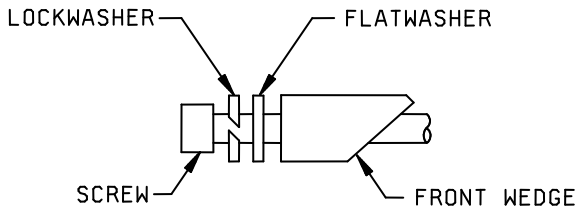



FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock vibration. Card holders requiring a screw self-locking element shall include configuration identifier "L" or "N" in the PIN (see classification, notes, and table I).

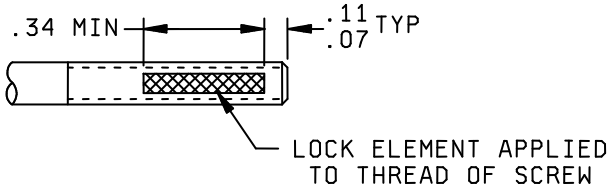



FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include configuration identifier "D" or "J" in the PIN (see classification, notes, and table I).



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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59789 – Holder, Electrical Card, Wedge Retainers, 5 Piece, For Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59789.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

| Manufacturer CAGE | Manufacturer name and address | Manufacturer contact information |
|---|---|---|
| 18915 | Birtcher – A Division of Pentair Electronic Packaging<br>7238 Trade Street<br>San Diego, CA 92121–3410 | Telephone: (858) 740–2400<br>Toll Free: (800) 854–7086<br>Facsimile: (858) 679–4555<br>(URL): www.birtcherproducts.com |
| 52094 | Calmark – A Division of Pentair Electronic Packaging<br>7238 Trade Street<br>San Diego, CA 92121–3410 | Telephone: (858) 740–2400<br>Toll Free: (800) 854–7086<br>Facsimile: (858) 740–2430<br>Electronic mail: sales@calmark.com<br>URL: www.calmark.com |
| 5BG68 | Card Locks Unlimited, Inc.<br>2310 E. Orangethorpe Avenue<br>Anaheim, CA 92806–1231 | Telephone: (714) 738–6194<br>Facsimile: (714) 446–0119<br>E-mail: sales@clumfg.com<br>URL: www.clumfg.com |

Part number supersession data. This CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "60". See table IV for CID PIN construction using other available lengths for this CID specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.



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TABLE III. Commercial part number supersession data.

| PIN designator AA59789/03 | Vendor similar designator or type part number 1/ CAGE 18915 | PIN designator AA59789/03 | Vendor similar designator or type part number 1/ CAGE 52094 | Vendor similar designator or type part number 1/ CAGE 5BG68 |
|---|---|---|---|---|
| AB60S | 42–5–12–B–LN | CB60S | A231CR5.80T2R | 5231BA–5.80T2R |
| AB60R | 42–5–12–B–LN–R | CB60R | A231CR5.80HR | 5231BA–5.80HR |
| AB60T | 42–5–12–B–LN–T | CB60T | A231CR5.80T0R | 5231BA–5.80T0R |
| AB60P | 42–5–12–B–LN–P | CB60U | A231CR5.80TM2R | 5231BA–5.80TM2R |
| FB60S | 42–5–12–B–LFLN | WB60S | WA231CR5.80T2R | 5231WBA–5.80T2R |
| FB60R | 42–5–12–BLFLNR | WB60R | WA231CR5.80HR | 5231WBA–5.80HR |
| FB60T | 42–5–12–BLFLNT | WB60T | WA231CR5.80T0R | 5231WBA–5.80T0R |
| FB60P | 42–5–12–BLFLNP | WB60U | WA231CR5.80TM2R | 5231WBA–5.80TM2R |
| NB60S | 42–5–12–B–LN–L | LB60S | A231CR5.80T2LR | 5231BA–5.80T2LR |
| NB60R | 42–5–12–B–LNRL | LB60R | A231CR5.80HLR | 5231BA–5.80HLR |
| NB60T | 42–5–12–B–LNTL | LB60T | A231CR5.80T0LR | 5231BA–5.80T0LR |
| NB60P | 42–5–12–B–LNPL | LB60U | A231CR5.80TM2LR | 5231BA–5.80TM2LR |
| JB60S | 42–5–12–BLFLNL | DB60S | WA231CR5.80T2LR | 5231WBA–5.80T2LR |
| JB60R | 42–5–12BLFLNR | DB60R | WA231CR5.80HLR | 5231WBA–5.80HLR |
| JB60T | 42–5–12BLFLNTL | DB60T | WA231CR5.80T0LR | 5231WBA–5.80T0LR |
| JB60P | 42–5–12BLFLNPL | DB60U | WA231CR5.80TM2LR | 5231WBA–5.80TM2LR |
| AC60S | 42–5–12–LN | CC60R | 231CR5.80T2R | 5231CG–5.80T2R |
| AC60R | 42–5–12–LN–R | CC60S | 231CR5.80HR | 5231CG–5.80HR |
| AC60T | 42–5–12–LN–T | CC60T | 231CR5.80T0R | 5231CG–5.80T0R |
| AC60P | 42–5–12–LN–P | CC60U | 231CR5.80TM2R | 5231CG–5.80TM2R |
| FC60S | 42–5–12–LF–LN | WC60S | W231CR5.80T2R | 5231WCG–5.80T2R |
| FC60R | 42–5–12–LF–LNR | WC60R | W231CR5.80HR | 5231WCG–5.80HR |
| FC60T | 42–5–12–LF–LNT | WC60T | W231CR5.80T0R | 5231WCG–5.80T0R |
| FC60P | 42–5–12–LF–LNP | WC60U | W231CR5.80TM2R | 5231WCG–5.80TM2R |
| NC60S | 42–5–12–LN–L | LC60S | 231CR5.80T2LR | 5231CG–5.80T2LR |
| NC60R | 42–5–12–LN–R–L | LC60R | 231CR5.80HLR | 5231CG–5.80HLR |
| NC60T | 42–5–12–LN–T–L | LC60T | 231CR5.80T0LR | 5231CG–5.80T0LR |
| NC60P | 42–5–12–LN–P–L | LC60U | 231CR5.80TM2.5LR | 5231CG–5.80TM2LR |

See footnote at end of table.



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TABLE III. Commercial part number supersession data – Continued.

| PIN designator AA59789/03 | Vendor similar designator or type part number 1/ CAGE 18915 | PIN designator AA59789/03 | Vendor similar designator or type part number 1/ CAGE 52094 | Vendor similar designator or type part number 1/ CAGE 5BG68 |
|---|---|---|---|---|
| JC60S | 42–5–12–LF–LNL | DC60S | W231CR5.80T2LR | 5231WCG–5.80T2LR |
| JC60R | 42–5–12–LFLNRL | DC60R | W231CR5.80HLR | 5231WCG–5.80HLR |
| JC60T | 42–5–12–LFLNTL | DC60T | W231CR5.80T0LR | 5231WCG–5.80T0LR |
| JC60P | 42–5–12–LFLNPL | DC60U | W231CR5.80TM2.5LR | 5231WCG–5.80TM2LR |
| AP60S | 42–5–12–N–LN | CE60S | EN231CR5.80T2R | 5231EN–5.80T2R |
| AP60R | 42–5–12–N–LN–R | CE60R | EN231CR5.80HR | 5231EN–5.80HR |
| AP60T | 42–5–12–N–LN–T | CE60T | EN231CR5.80T0R | 5231EN–5.80T0R |
| AP60P | 42–5–12–N–LN–P | CE60U | EN231CR5.80TM2R | 5231EN–5.80TM2R |
| FP60S | 42–5–12–N–LFLN | WE60S | WEN231CR5.80T2R | 5231WEN–5.80T2R |
| FP60R | 42–5–12–NLFLNR | WE60R | WEN231CR5.80HR | 5231WEN–5.80HR |
| FP60T | 42–5–12–NLFLNT | WE60T | WEN231CR5.80T0R | 5231WEN–5.80T0R |
| FP60P | 42–5–12–NLFLNP | WE60U | WEN231CR5.80TM2R | 5231WEN–5.80TM2R |
| NP60S | 42–5–12–N–LN–L | LE60S | EN231CR5.80T2LR | 5231EN–5.80T2LR |
| NP60R | 42–5–12–N–LNRL | LE60R | EN231CR5.80HLR | 5231EN–5.80HLR |
| NP60T | 42–5–12–N–LNTL | LE60T | EN231CR5.80T0LR | 5231EN–5.80T0LR |
| NP60P | 42–5–12–N–LNPL | LE60U | EN231CR5.80TM2LR | 5231EN–5.80TM2LR |
| JP60S | 42–5–12–NLFLNL | DH60S | WEN231CR5.80T2LR | 5231WEN–5.80T2LR |
| JP60R | 42–5–12NLFLNRL | DH60R | WEN231CR5.80HLR | 5231WEN–5.80HLR |
| JP60T | 42–5–12NLFLNTL | DH60T | WEN231CR5.80T0LR | 5231WEN–5.80T0LR |
| JP60P | 42–5–12NLFLNPL | DH60U | WEN231CR5.80TM2LR | 5231WEN–5.80TM2LR |
| AH60S | 42–5–12–B3–LN | CH60S | HA231CR5.80T2R | 5231BH–5.80T2R |
| AH60R | 42–5–12–B3–LNR | CH60R | HA231CR5.80HR | 5231BH–5.80HR |
| AH60T | 42–5–12–B3–LNT | CH60T | HA231CR5.80T0R | 5213BH–5.80T0R |
| AH60P | 42–5–12–B3–LNP | CH60U | HA231CR5.80TM2R | 5231BH–5.80TM2R |
| FH60S | 42–5–12–B3LFLN | WH60S | WHA231CR5.80T2R | 5231WBH–5.80T2R |
| FH60R | 42–5–12B3LFLNR | WH60R | WHA231CR5.80HR | 5231WBH–5.80HR |
| FH60T | 42–5–12B3LFLNT | WH60T | WHA231CR5.80T0R | 5231WBH–5.80T0R |
| FH60P | 42–5–12B3LFLNP | WH60U | WHA231CR5.80TM2R | 5231WBH–5.80TM2R |

See footnote at end of table.



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TABLE III. Commercial part number supersession data – Continued.

| PIN designator AA59789/03 | Vendor similar designator or type part number 1/ CAGE 18915 | PIN designator AA59789/03 | Vendor similar designator or type part number 1/ CAGE 52094 | Vendor similar designator or type part number 1/ CAGE 5BG68 |
|---|---|---|---|---|
| NH60S | 42–5–12–B3–LNL | LH60S | HA231CR5.80T2LR | 5231BH–5.80T2LR |
| NH60R | 42–5–12–B3LNRL | LH60R | HA231CR5.80HLR | 5231BH–5.80HLR |
| NH60T | 42–5–12–B3LNTL | LH60T | HA231CR5.80T0LR | 5231BH–5.80T0LR |
| NH60P | 42–5–12–B3LNPL | LH60U | HA231CR5.80TM2LR | 5231BH–5.80TM2LR |
| JH60S | 42–5–12B3LFLNL | DH60S | WHA231CR5.80T2LR | 5231WBH–5.80T2LR |
| JH60R | 42–512B3LFLNRL | DH60R | WHA231CR5.80HLR | 5231WBH–5.80HLR |
| JH60T | 42–512B3LFLNTL | DH60T | WHA231CR5.80T0LR | 5231WBH–5.80T0LR |
| JH60P | 42–512B3LFLNPL | DH60U | WHA231CR5.80TM2LR | 5231WBH–5.80TM2LR |

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.

TABLE IV. Example of PIN with available length designators.

| PIN designator AA59789/03 | Vendor similar designator or type part number 1/ 2/ CAGE 18915 | PIN designator AA59789/03 | Vendor similar designator or type part number 1/ 2/ CAGE 52094 | Vendor similar designator or type part number 1/ 2/ CAGE 5BG68 |
|---|---|---|---|---|
| NH50T | 42–5–10–B3LNTL | LH50T | HA231CR4.80T0LR | 5231BH–4.80T0LR |
| NH55T | 42–5–11–B3LNTL | LH55T | HA231CR5.30T0LR | 5231BH–5.30T0LR |
| NH60T | 42–5–12–B3LNTL | LH60T | HA231CR5.80T0LR | 5231BH–5.80T0LR |
| NH65T | 42–5–13–B3LNTL | LH65T | HA231CR6.30T0LR | 5231BH–6.30T0LR |

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.

2/ Other lengths are available on request.



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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

Project 5998–2010–003

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